

Predicting NBA Player Positions Using Statistical Profiles

Using per-75 possession statistical profiles, KNN, SVM, K-means, and hierarchical clustering

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NBA
analytics



Project question & storyline

What we tried to learn from player statistics

RESEARCH QUESTION

Can NBA player statistical profiles accurately predict whether a player is a guard or non-guard?

BASKETBALL POSITIONS & RESPONSIBILITIES				
 POINT GUARD	 SHOOTING GUARD	 SMALL FORWARD	 POWER FORWARD	 CENTER
PG	SG	SF	PF	C
• Runs the offense • Distributes the ball • Maintains court vision • Directs traffic • Controls game tempo	• Scores points • Shoots from long range • Slashes to the basket • Best shooter • Defends perimeter	• Balanced role • Scores & defends • Rebounds • Matches up with opponent • Adapts to situations	• Scores in the paint • Sets screens • Crashes the boards • Provides physicality • Strong rebounder	• Protects the rim • Controls the paint • Rebounds • Sets hard screens • Tallest player

Secondary Question:

Do players naturally group into clusters based on their statistical profiles?



Rationale for PER 75

Standardizing player performance across contexts

- Per-game stats are influenced by pace and playing time
- Fast teams and high-minute players appear inflated
- Per-75 possessions standardizes production
- Allows fair comparison across players, roles, and eras, which is critical for building reliable machine learning models.

PER GAME

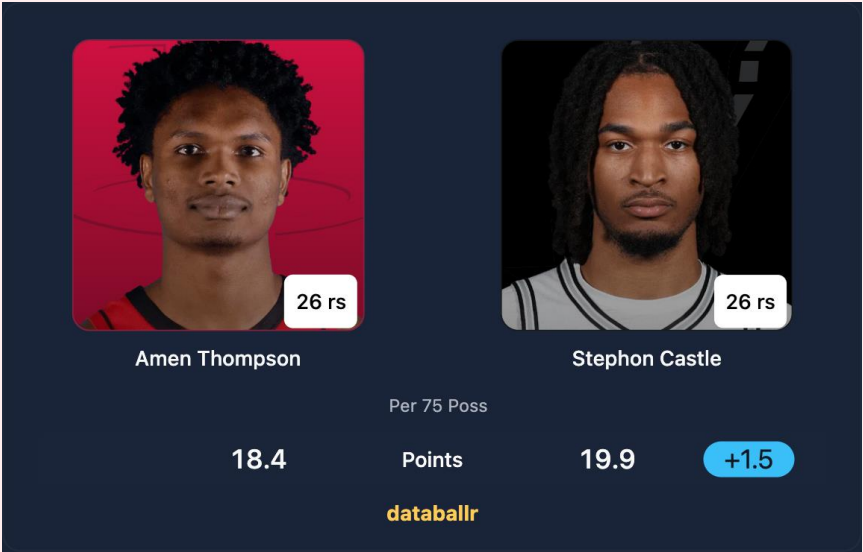


Minutes per game

Castle: 30.0 MPG

Amen: 37.4 MPG
(1st in NBA)

PER 75



Data Source

Where the dataset comes from and how it was constructed

Source: Stathead Basketball



- Data Filters Applied:
 - Seasons: 1981–82 through 2025–26
 - Regular Season only
 - Minimum Games Played ≥ 42
 - Ranked by Points per 75 Possessions
 - Top 600 single-season performances

Why these filters?

- Ensuring data quality and fair comparison
 - Start from 1981–82: first 3-point era season with complete data (no missing values)
 - Regular season only: avoids playoff variability and keeps comparisons consistent
 - Minimum 42 games played:
 - Ensures reliable sample size ($>$ half of 82-game season)
 - Avoids small-sample bias
 - Still includes shortened seasons (e.g., COVID years)

Data overview

NBA player seasons ranked by scoring profile

600

player-season observations

33

original variables

28

numeric predictors used

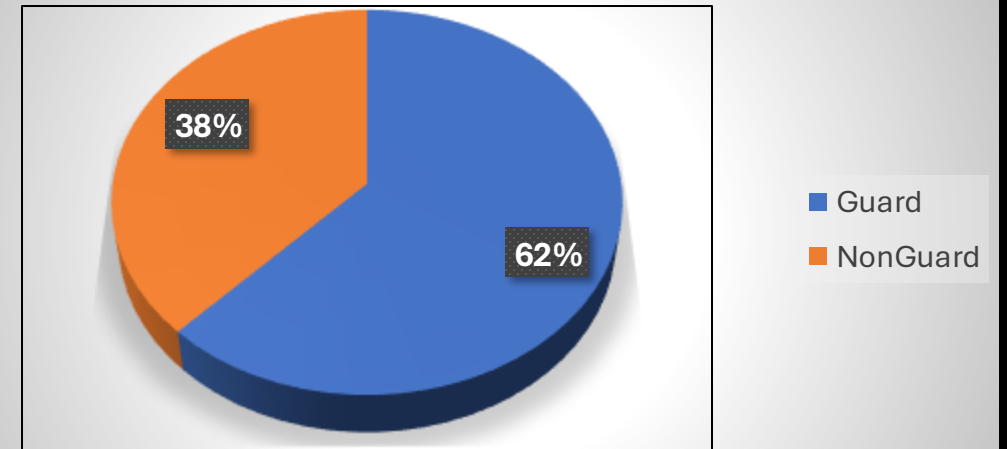
2

classes: Guard / NonGuard

DATA FIELDS

- Categorical: Player, Season, Team, Position
- Numerical: Age, G, GS, MP, AS
- All box score stats are per 75 possessions
- Includes shooting, rebounding, and playmaking metrics
- Key variables: PTS/75, PTS/G, efficiency (FG%, TS%, eFG%)
- Per-75 allows fair comparison across players and eras

CLASS BALANCE



Preprocessing decisions

Preparing data for distance-based and kernel methods

Step	Why it matters
Removed Player, Team, Season	Identifiers do not contribute to prediction
Created GuardClass	Defines classification target
Replaced 3P% NA → 0	Players with no 3PA have undefined 3P%; 0 keeps all 600 observations
Centered and scaled predictors	Needed because KNN, SVM, and clustering rely on distances
Retained numeric features	Preserves full statistical profile and basketball interpretability

These steps ensure fair distance calculations and improve model reliability.

Why Scaling Matters

Ensuring fair contribution of all variables

Models like KNN, SVM, and clustering rely on distance calculations

Without scaling:

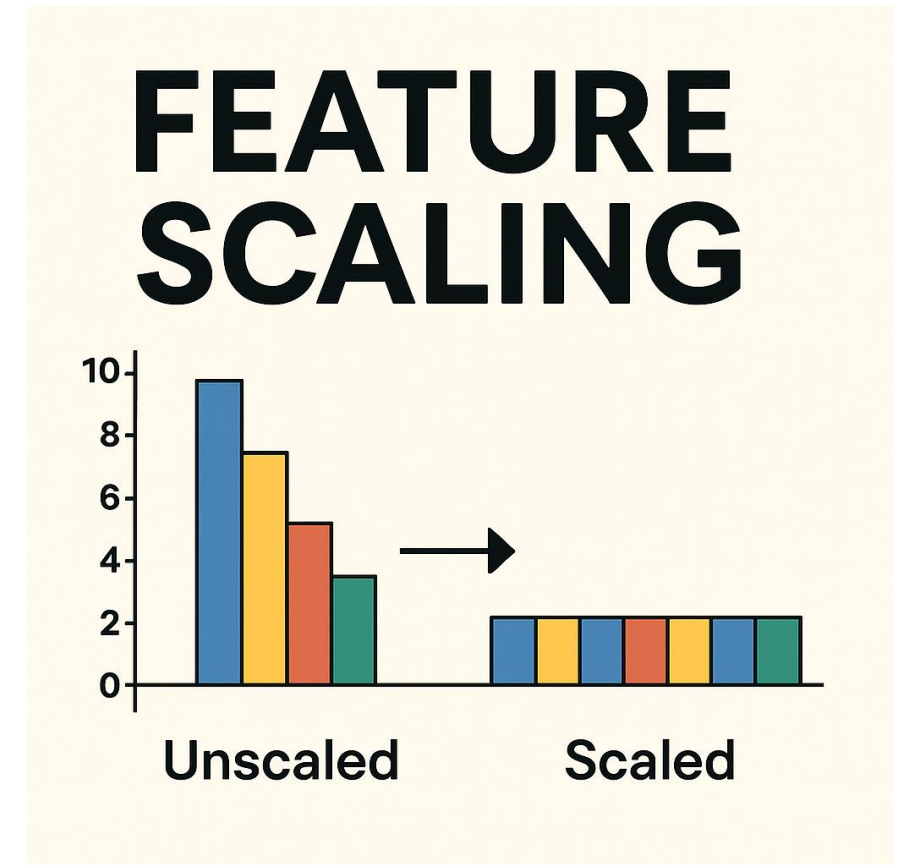
- Variables with large values dominate

With scaling:

- All features contribute equally

Supervised models use labels, clustering does not

This step is critical or results become meaningless



Modeling plan

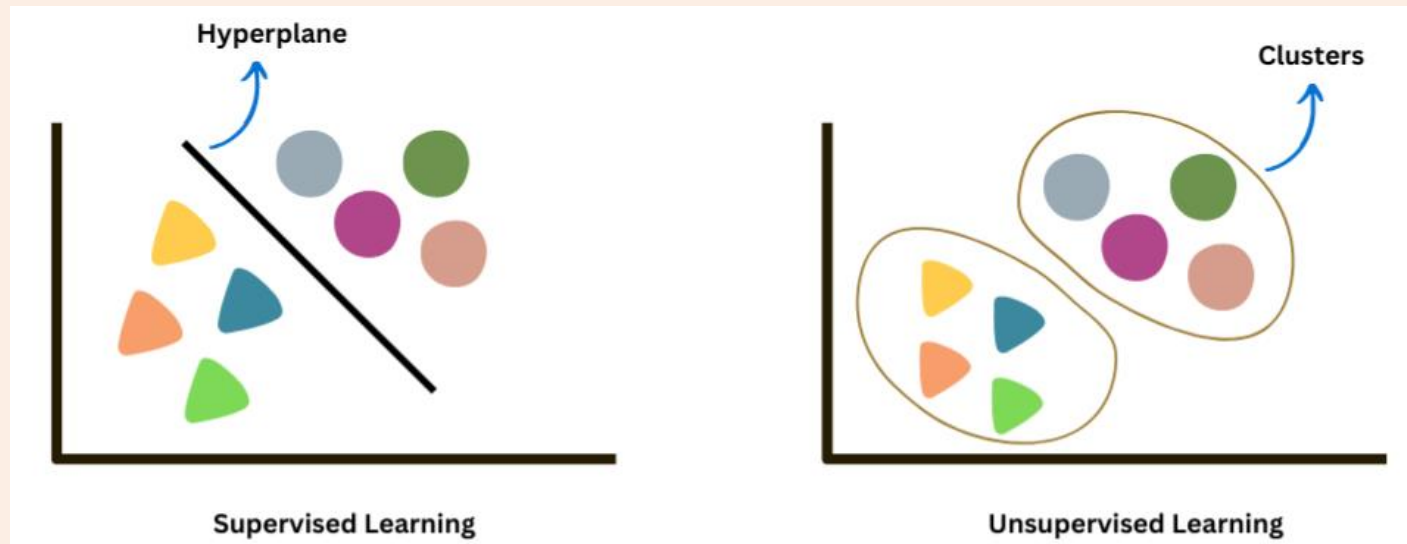
Two supervised models + two clustering models

Supervised classification

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

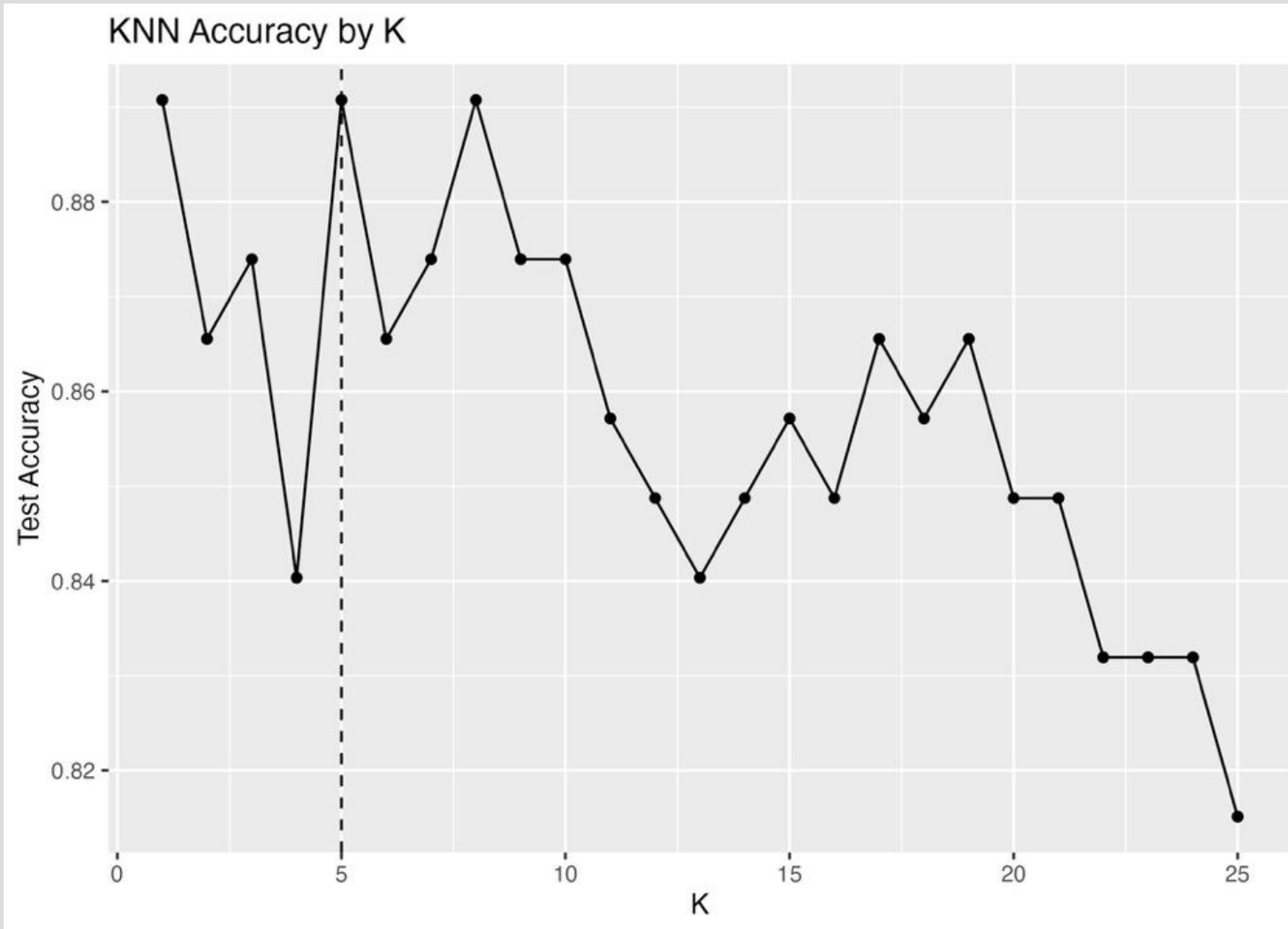
Unsupervised clustering

- K-Means Clustering
- Hierarchical Clustering



KNN results

Choosing K from test accuracy



KNN summary

- K tested: 1 to 25
- Chosen K: 5
- Test accuracy: 89.08%
- Balanced accuracy: 87.30%

K = 5 was selected because it tied for best accuracy while avoiding the instability of K = 1.

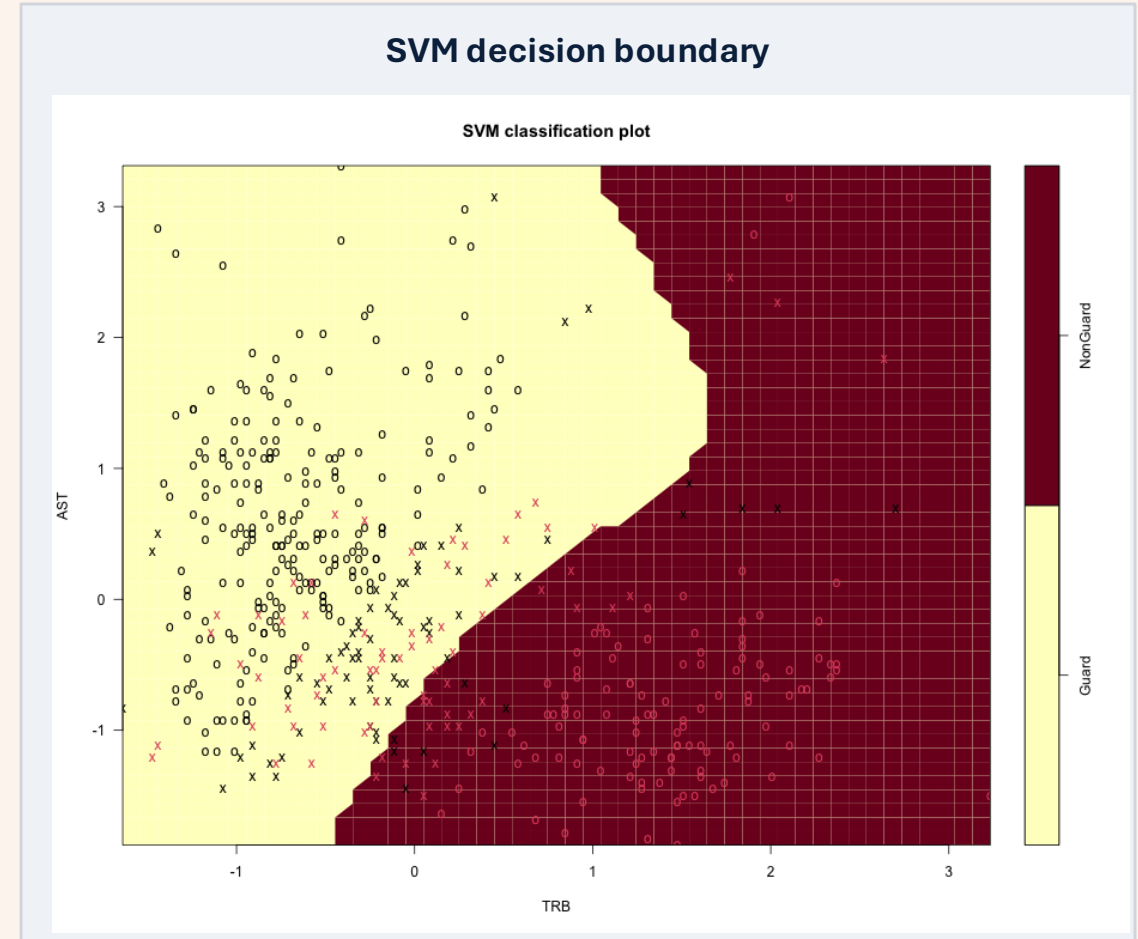
SVM results

Best supervised model using an RBF kernel

Tuned using cross-validation

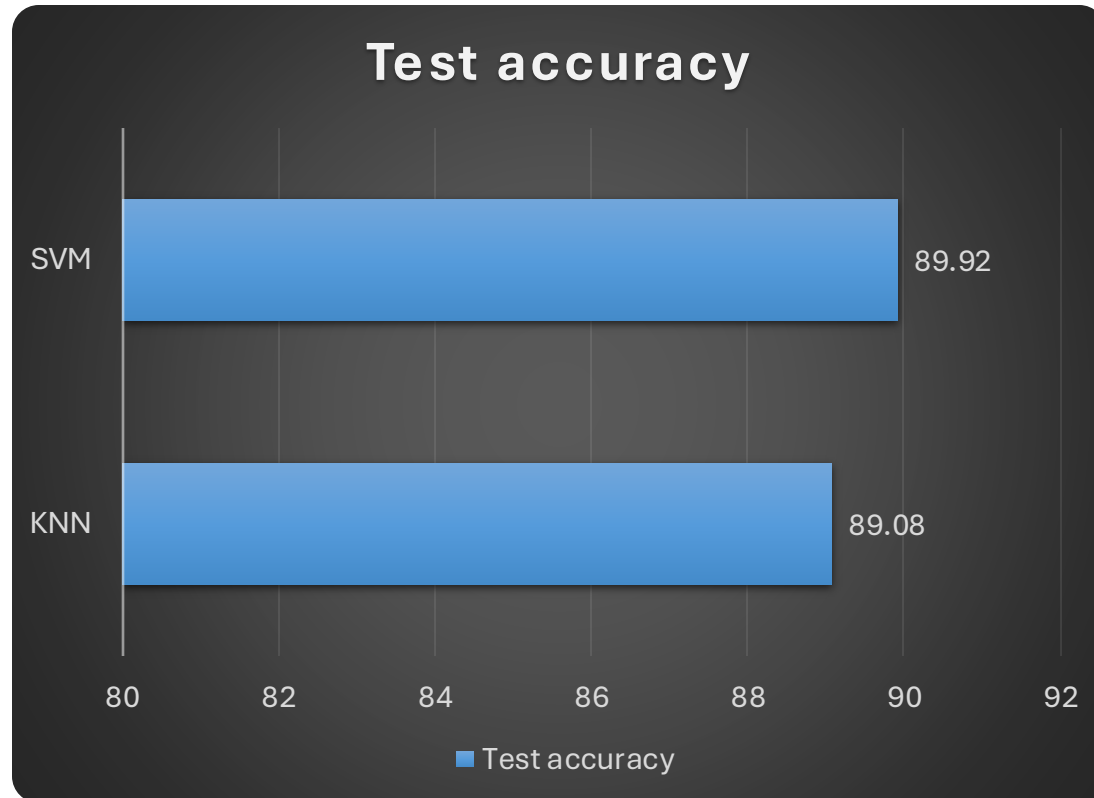
- Best parameters:
 - Cost = 10
 - Gamma = 0.1
- Test accuracy: 89.92%

The SVM slightly outperformed KNN



Supervised model comparison

Both models predict position well; SVM is slightly better



CONFUSION MATRICES

KNN	Actual G	Actual NG
Pred G	70	9
Pred NG	4	36

SVM	Actual G	Actual NG
Pred G	70	8
Pred NG	4	37

Takeaway: SVM reduced one additional non-guard error compared with KNN, giving it the best test accuracy.

Clustering setup

Testing whether natural groups match basketball roles

K-means

- Tested K = 2 to 10
- Best K = 2
- Silhouette Score = 0.174
 - Indicates weak cluster separation
- Goal: compact groups of similar players

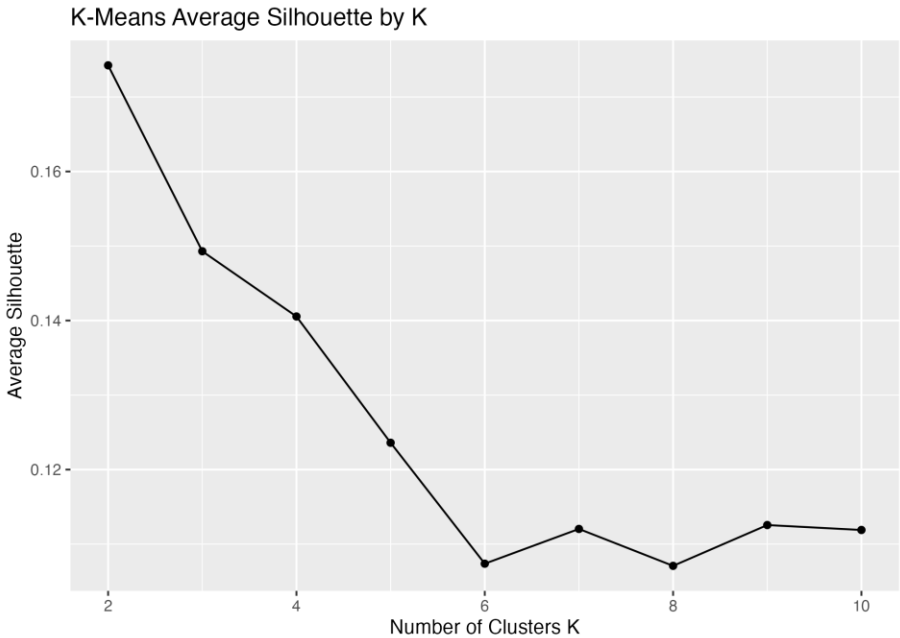
Hierarchical

- Tested complete, average, and Ward linkage
- Selected average linkage
- Best K = 2
- Silhouette Score = 0.208
 - Slightly better than K-Means
- Goal: visualize cluster hierarchy

Silhouette values closer to 1 indicate cleaner separation; values near 0 suggest overlap.

K-means clustering results

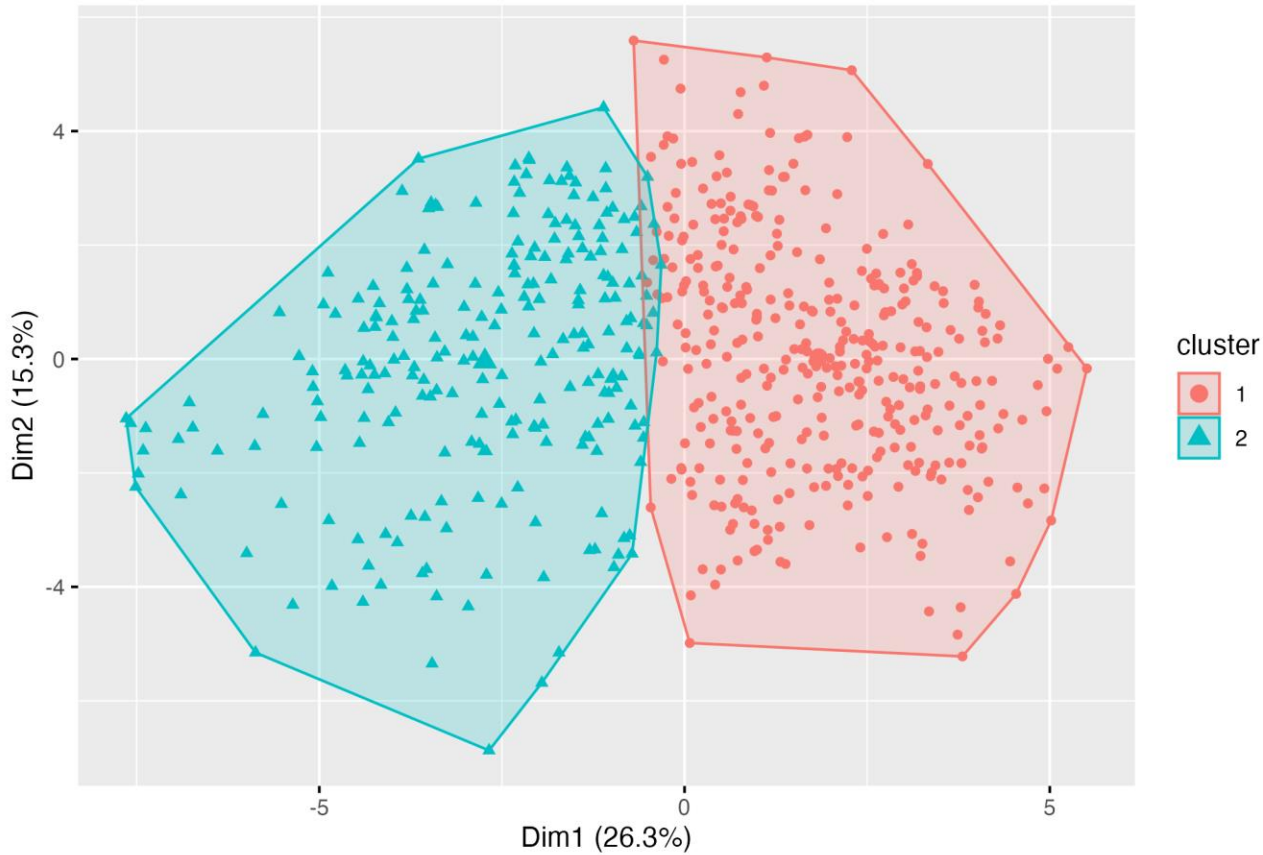
K = 2 had the best silhouette score



MATRIX

K-means	Guard	NonGuard
Cluster 1	295	64
Cluster 2	78	163

K-Means Clustering of NBA Player Profiles



Interpretation

- Cluster 1 is guard-heavy.
- Cluster 2 has more non-guards.
- Separation exists, but overlap is still meaningful.

Hierarchical clustering results

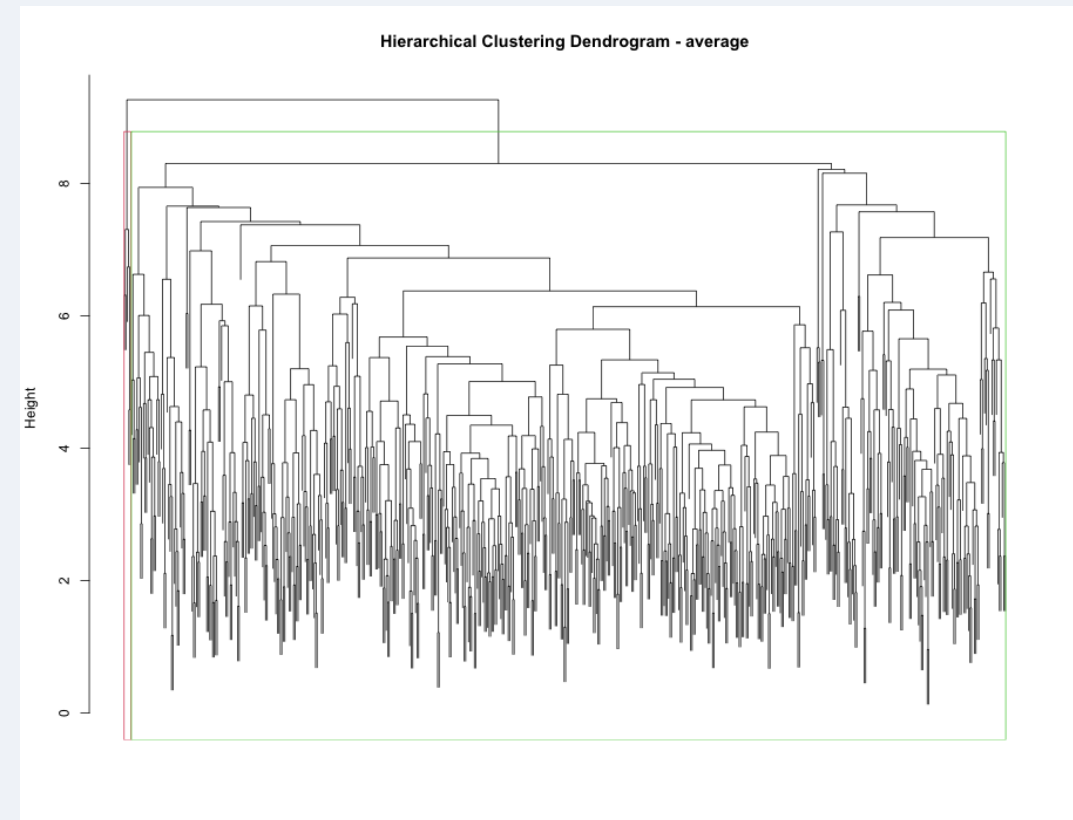
Average linkage performed best

Best configuration

- Linkage: average
- $K = 2$
- Silhouette: 0.208
- Slightly higher than K-means

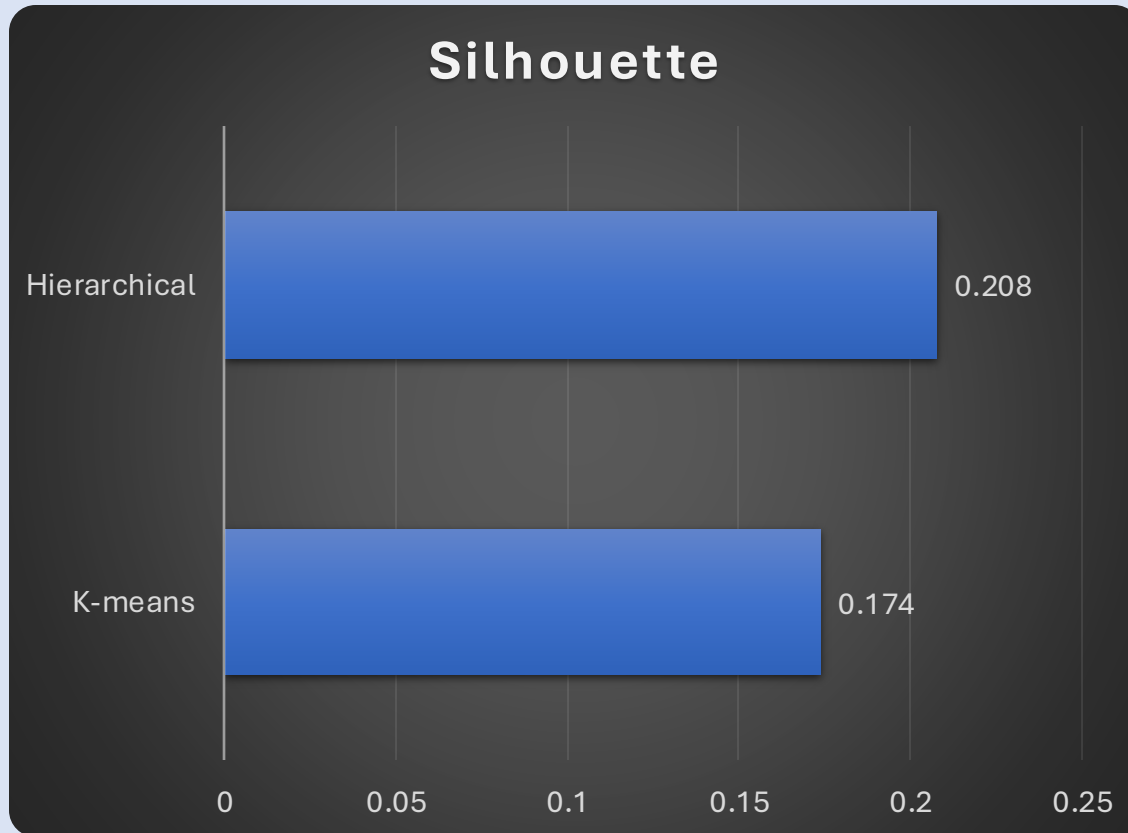
Hierarchical	Guard	NonGuard
Cluster 1	368	227
Cluster 2	5	0

Hierarchical dendrogram



Clustering comparison

Natural groups are weaker than supervised labels



Interpretation

- Hierarchical clustering performed slightly better than K-Means based on silhouette score.
- However, both methods produced relatively low scores, indicating weak cluster separation.
- This suggests that player positions are not clearly distinguishable based on statistical profiles alone.
- Significant overlap exists between guards and non-guards, reflecting shared skill sets.
- Overall, the data shows weak natural grouping, making clustering a challenging task.

Positional Flexibility in Modern Basketball

Why player roles are no longer clearly separated

Modern basketball has shifted toward greater positional flexibility.

Traditionally:

- Guards handled ball movement and perimeter shooting
- Forwards/Centers focused on rebounding and interior play

Today:

- Players often perform multiple roles (e.g., point-forwards, stretch bigs)
- Skill sets overlap across positions

As a result:

- Statistical profiles between guards and non-guards are less distinct
- Clear boundaries between positions are harder to define

This helps explain why clustering methods showed weak separation and overlapping groups in our results.



Final model on the full dataset

SVM was selected as the best supervised model

382

support vectors

209 / 173

Guard / NonGuard support vectors

100%

full-data accuracy

Important caution

The 100% full-data accuracy is not the true predictive estimate because the model is evaluated on the same data it was trained on. The realistic performance estimate is the test accuracy: 89.92%.

Limitations and difficulties

What made the analysis challenging

- Hybrid positions such as G-F and F-G blur class boundaries.
- Correlated efficiency variables such as TS% and eFG% may add redundancy.
- Clustering had low silhouette scores, meaning natural separation was weak.
- The data captures statistical profile, but not context such as team role or defensive assignments.

Key Insight

Position labels are predictable, but player roles are more continuous than discrete.

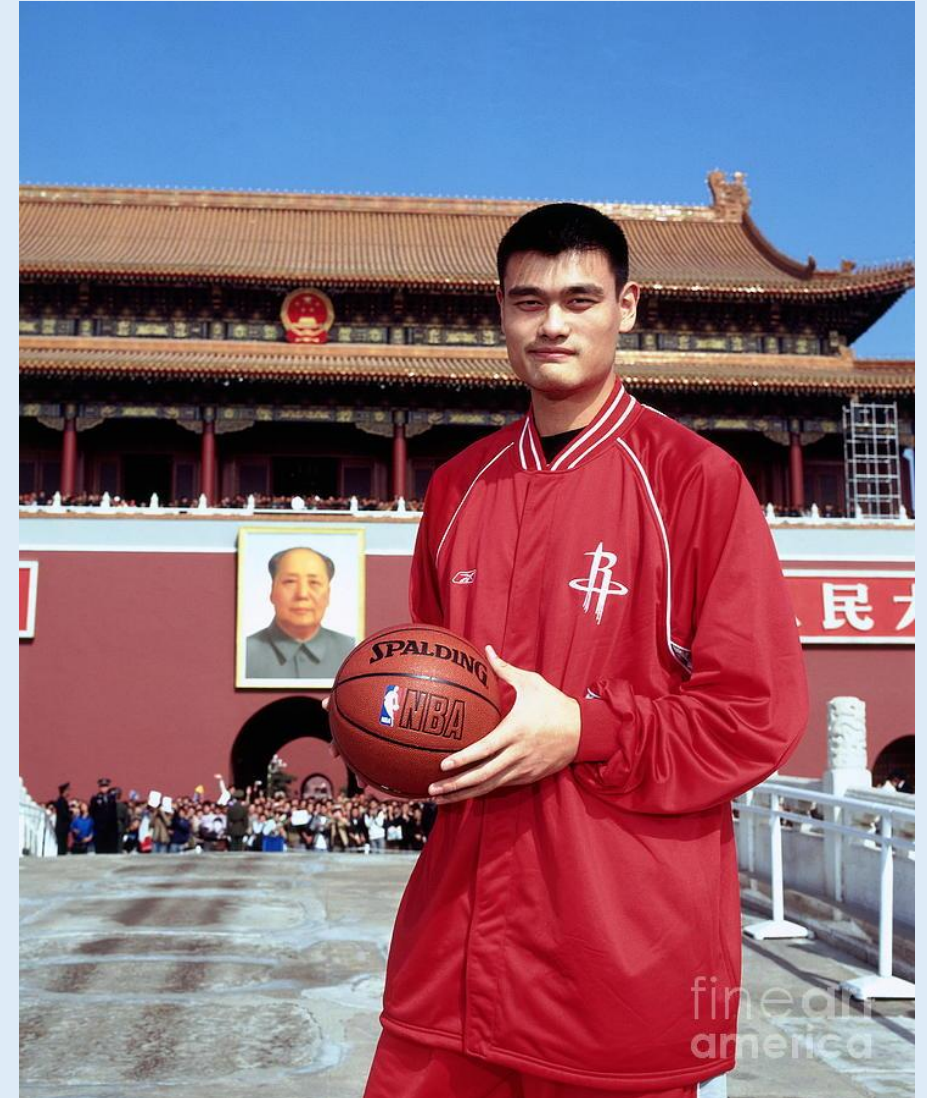


Possible improvements

How the analysis could be extended

Several improvements could strengthen both predictive performance and interpretability:

- Feature Engineering
 - Apply feature selection or PCA to reduce redundancy (e.g., TS% vs eFG%)
 - Improve model interpretability and reduce noise
- Add More Context
 - Include role-based variables (usage rate, lineup role)
 - Add defensive metrics and team system information
 - Capture factors not reflected in box score stats
- More Advanced Models
 - Use Random Forests or Neural Networks
 - Capture non-linear interactions between features
- Improved Clustering Methods
 - Apply DBSCAN or model-based clustering
 - Better handle overlapping and uneven player groups



Conclusion

Main answer:

NBA statistical profiles can predict guard vs non-guard position with about 90% test accuracy.

But clustering shows that player roles overlap, reflecting the positional flexibility of modern basketball.

Questions?



References

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